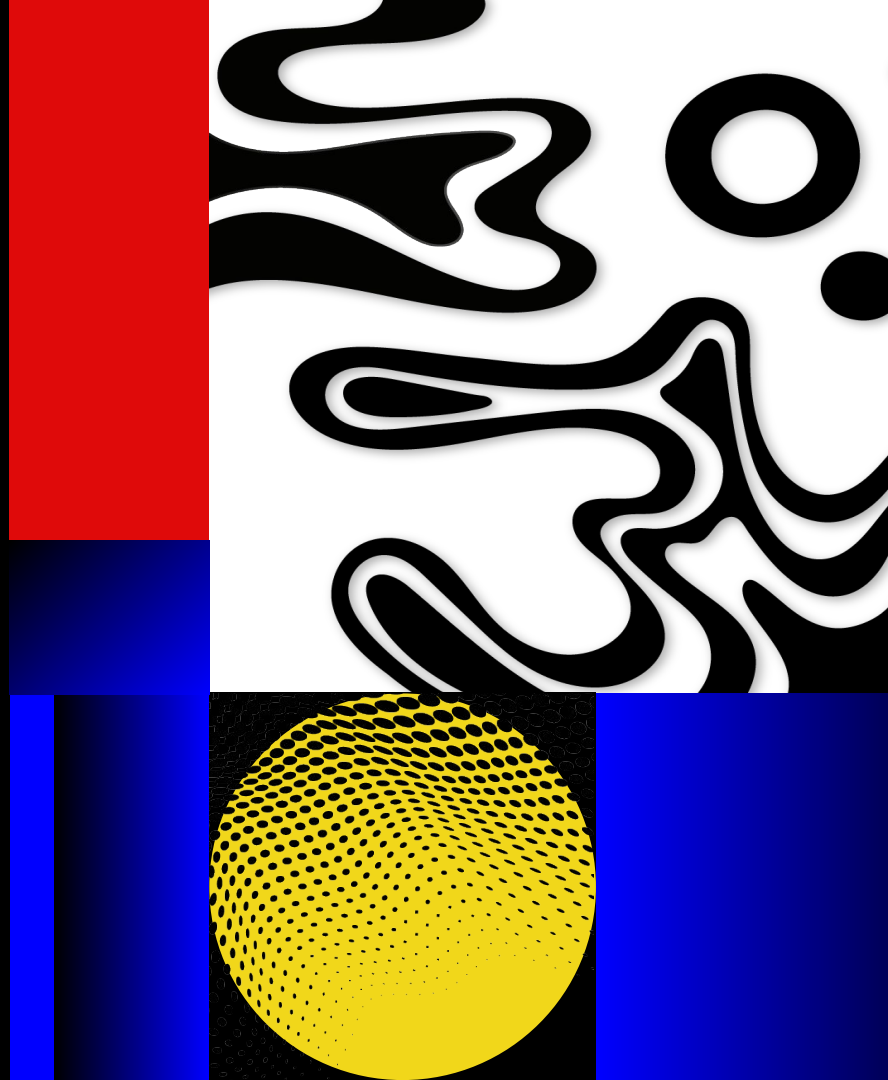
PAINTING FROM PART

Iulia Popa

Dongsheng Guo, Haoru Zhao, Yunhao Cheng,
Haiyong Zheng, Zhaorui Gu, Bing Zheng



CONTENTS

01 **Part-painting Problem and solution** — Brief description of part-painting

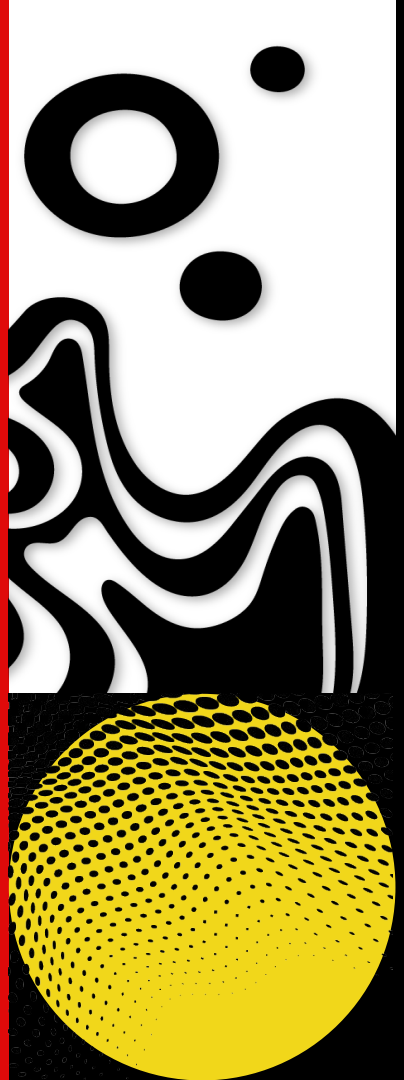
02 **Phases of Part-Painting**
- Part-noise Restarting
- Part-feature Repainting
- Part-patch Refining

03 **Related Work**
- Inpainting & Outpainting
- Generative Adversarial Networks
- Wavelet-based Generation Network
- Image-to-Image Translation



INTRODUCTION

Painting from Part (*Part-painting* for short) represents the procedure of painting the complete image from a part of it, using a simple human ability: given a region of a picture, people can visualize the unseen rest of the image.



01

Part-painting Problem and solution

The connection between *part* and *whole*

Painting from Part problem - real life applicability -



Computer Science

Computer vision
Texture analysis
Image-editing
Video games
Generated media



Visual Arts

Abstract art
Comics
Animated movies
Cartoons
Books

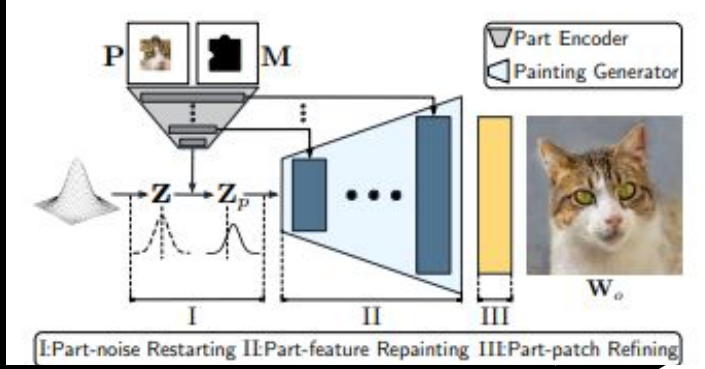
Painting from Part solution

- Ability to generate complete images from irregular shaped parts
- Utilization of the given part, a mask and random noise from a standard normal distribution
- Novel network architecture based On Generative Adversarial Networks

Phases of Part-Painting

02

Steps of the process



Part-noise Restarting

The system is restarted in the beginning in order to adjust normally distributed part-noise.

Part-feature Repainting

The process repaints the generated characteristics using feature transferring methods.

Part-patch Refining

The application makes the resulted image clearer by connecting the similar parts.

Part-noise Restarting

- Painting from noise instead of straight painting from given part
- Using the normal distribution of the local domain (input component)
- Reviewing the noise with the specifical encoder
- Utilizing the Hadamard product in formula
- Constructing distributions for each image for painting improvement

Part-feature Repainting

- Imitating the human's ability to complete a region by adding unseen elements
- Using a novel component, called *Repainting Residual Block (RRB)*
- Processing each layer, maintaing a strong connection between them
- Transferring procedures
- Reconstructing the whole-feature with convolutional methods

Part-patch Refining



Adjusting the whole-feature



Changing parts in pixels



Part-patch transmitting



Whole-patch fusioning



Returning the complete image as a result


Part
w/o RL
w/o FT
w/o FR
SN
TN
w/ RL
GT


Part
w/o PRM
PRM$_{feat}$
PRM$_{mid}$
PRM$_{L1}$
PRM$_{L2}$
Atten
w/ PRM
GT







03

Related Work

Other part-painting methods

1 Inpainting and Outpainting

2 Generative Adversarial Networks

3 Generative Image Inpainting with Auxiliary Contextual Reconstruction



4 Parallel Multi-Resolution Fusion Network

5 Wavelet-based Generation Network

6 Conditional Texture and Structure Dual Generation

7 Distillation-Guided Image Inpainting

8 Image-to-Image Translation



THANKS!

Do you have any questions?

References

[1] Dongsheng Guo, Haoru Zhao, Yunhao Cheng, Haiyong Zheng, Zhaorui Gu, Bing Zheng, Underwater Vision Lab, Sanya Oceanographic Institution, “Painting from Part”, pp. 14779-14786, ICCV 2021
[2] James Vincent, “This AI art app is a glimpse at the future of synthetic media”, The Verge publication, 2021
[3] J Yunjey Choi, Youngjung Uh, Jaejun Yoo, Jung-Woo Ha, “StarGAN v2: Diverse image synthesis for multiple domains”, CVPR, pp. 8188–8197, 2020
[4] Gao Huang, Zhuang Liu, Laurens Van Der Maaten, Kilian Q Weinberger, “Densely connected convolutional networks”, CVPR, pp. 4700–4708, 2017
[5] Philip Isola, Jun-Yan Zhu, Tinghui Zhou, Alexei A Efros, “Image-toimage translation with conditional adversarial networks”, CVPR, pp. 1125–1134, 2017
[6] Guilin Liu, Fitsum A Reda, Kevin J Shih, Ting-Chun Wang, Andrew Tao, Bryan Catanzaro, “Image inpainting for ir- regular holes using partial convolutions”, ECCV, pp. 85–100, 2018
[7] Takeru Miyato, Toshiki Kataoka, Masanori Koyama, Yuichi Yoshida, “Spectral normalization for generative adversarial networks”, ICLR, pp. 73-82, 2018
[8] Yu Zheng, Zhe Lin, Huchuan Lu, Vishal M Patel, “CR-Fill: Generative Image Inpainting With Auxiliary Contextual Reconstruction”, ICCV, pp. 14164-14173, 2021
[9] Wentao Wang, Jianfu Zhang, Li Niu, Haoyu Ling, Xue Yang, Liqing Zhang, “Parallel Multi-Resolution Fusion Network for Image Inpainting”, ICCV, pp. 14559-14568, 2021
[10] Yingchen Yu, Fangneng Zhan, Shijian Lu, Jianxiong Pan, Feiying Ma, Xuansong Xie, Chunyan Miao, “WaveFill: A Wavelet-Based Generation Network for Image Inpainting”, ICCV, pp. 14114-14123, 2021
[11] Maitreya Suin, Kuldeep Purohit, A N Rajagopalan, “Distillation-Guided Image Inpainting”, ICCV, pp. 2481-2490, 2021
[12] Xiefan Guo, Hongyu Yang, Di Huang, “Image Inpainting via Conditional Texture and Structure Dual Generation”, ICCV, pp. 14134-14143, 2021